Severing verbs from their arguments:
Directionals and argument structure in Mam[1]

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Abstract

This paper proposes an analysis of argument structure in Mam, with a focus on directional auxiliary verbs (henceforth directionals). We argue that directionals play a fundamental role for the derivation of argument structure that goes beyond their well-known contribution to spatio-temporal deixis. The evidence for this proposal comes from the observation that transitive and unaccusative verbs *require* co-occurrence with a directional, whereas unergative verbs never do. We propose a decompositional analysis of argument structure in event semantics (Davidson 1967), arguing for a neo-Davidsonian approach in which Mam verbs ought to be severed not just from their external arguments (à la Schein 1993 and Kratzer 1996), but also from their internal arguments. The result is a theory in which verbs constitute mere properties of events (as in e.g., Castañeda 1967, Borer 2005, Pietroski 2005, Levinson 2010, Williams 2009, Londahl 2014, Elliott 2020), and so directionals are required to introduce themes. Finally, we provide a reinterpretation of Mam's exceptionally rich verbal morphology, proposing specifically that the inventory of voice-related functional heads includes both agentive and (several flavors of) passive morphology. We then argue that these heads provide us with overt evidence for the existence of argument-introducing verbal functional heads (as in e.g., Kratzer 1996, Harley 2014), and point to the possibility that at least some passive morphemes may be involved, in addition to directionals in Mam, in the introduction of theme arguments.

Word count: 16,891 (including abstract, footnotes, and references)

[1]Abbreviations in this paper follow Leipzig glosses with the following additions: A – "Set A" (ergative, possessive); AP – antipassive suffix; B – "Set B" (absolutive); COM – completive aspect; DA – disagreement suffix; DIR – directional; DS – directional suffix; MW – measure word; NF – nonfinte suffix; PAT – patient; PROX – proximate aspect; RN – relational noun; TV – transitive voice suffix; v.AGT – agentive v morpheme. In some cases glosses from other works have been adapted for consistency.

1 Introduction

Many Mayan languages employ directional auxiliary verbs (henceforth "directionals") in the expression of verbal or stative predicates (see e.g., England 1976, 2001; Martin 1977; Haviland 1993; Zavala 1993; Aissen 1994; Mateo Toledo 2004; 2008, 2012, 2023; Henderson 2016; Aissen, England, and Maldonado 2017; Henderson, Elias, Royer, and Coon 2018; Elias 2019). Among other uses discussed below, these items are best known for contributing information pertaining to the direction, location, or time of an event. A near minimal pair from the Mam dialect of Todos Santos Cuchumatán is provided below.

(1) a. Ma ∅ **pon** t-ii-'n Noé kiẍ
PROX B2/3SG DIR:arrive.there A2/3SG-bring-DS Noé fish
'Noé took the fish there.' (TS Mam)

b. Ma tz=**ul** t-ii-'n Noé kiẍ
PROX B2/3SG=DIR:arrive.here A2/3SG-bring-DS Noé fish
'Noé brought the fish here.' (TS Mam)

In (1a), the directional *pon* combines with the verbal root *ii* 'to bring' to convey that the agent, *Noé*, brought the fish to a location away from the speaker. With the directional *ul* in (1b), on the other hand, it is understood that the fish was brought towards the speaker.[2]

The main goal of this paper is to argue that directionals in Mam have functions that go beyond that of deixis, namely playing a fundamental role for argument structure. This echoes other work on Mayan languages, which have argued that directionals are needed for different reasons broadly relating to argument structure (Mateo Toledo 2008, 2023, Henderson et al. 2018, Elias 2019). Here, we specifically argue that directionals are needed in Mam to introduce theme arguments, showing that the following empirical generalizations hold in at least three dialects of Mam.

(2) Generalizations about directionals in Mam:
a. All (but one) transitive verbs require a directional;
b. All (underived) unaccusative verbs require a directional;
c. Unergative verbs never require a directional;
d. Voice derivations (antipassive and passive) circumvent the need for directionals.

Some analyses of transitive and unaccusative verbal roots in languages like English (Kratzer 1996, Harley 2014), or even in Mayan languages (see e.g., Coon 2019, Coon and Royer 2020), have proposed that they are lexically specified to select for an internal argument. Since Kratzer 1996, it is often assumed that internal arguments (generally themes) differ from external arguments (generally agents) in that way. Namely, external arguments are proposed to be *severed* from the argument structure of verbal roots. With this in mind, the generalizations in (2a-2c), taken together, point to a natural hypothesis about the need for directionals in Mam. If internal arguments are in fact severed from the argument structure of verbal roots (as external arguments have been taken to be in works like Kratzer 1996), then perhaps directionals instantiate the ingredient needed to introduce them, rendering Mam verbal roots argumentless. This is exactly what we propose:

[2] While it is beyond the scope of this paper to discuss the behavior of Mam directionals within a general framework of 'associated motion' constructions common in Mesoamerican languages, we refer the interested reader to recent overviews in Guillaume and Koch 2021.

(3) Main theoretical proposals about Mam argument structure:

a. Verbal roots denote properties of events, e.g.: $[\![feed]\!] = \lambda e.\text{FEED}(e)$;

b. Themes (in active sentences) are introduced by a functional head, "DIR," headed by the directional auxiliary verb;

c. Agents are introduced by a functional head, *v*/Voice, lexically specified as *-n*.

This paper is structured as follows. Section 2 provides basic morphosyntactic background on Mam, as well as information about data and methodology. Section 3 then proposes to disentangle two uses of directionals: "lexical uses," describing their deictic contribution; and "functional uses," describing their relation to theme introduction. Section 4 zooms in on the functional use of directionals, providing data supporting the generalizations in (2a-2c). Section 5 then turns to the components of the analysis in (3a-3b), and also discusses the generalization in (2d). As we will show, (2d) not only provides evidence for our analysis, but also leads to novel insight for the syntactic analysis of passives in Mam: that a subset of passive morphemes are also involved in the introduction of themes. Section 6 then zooms into the introduction of external arguments, component (3c) of our proposal. We argue that external arguments must also be severed from the argument structure of verbal roots, and propose that the introduction of agents—just like that of themes—is signaled by overt morphology in the language, in this case *-n*. That section will also show how our account can provide a comprehensive approach to word and morpheme order in Mam. Section 7 concludes.

2 Basics of Mam syntax

Mam (iso 639: `mam`) is a Mamean-branch Mayan language spoken predominantly in Guatemala, but also in large diaspora communities across North America, by over 500,000 speakers (Richards and Macario 2003). The data in this paper come from several varieties, either by means of data collection by one of the authors (using a hypothesis-driven fieldwork methodology; e.g., Matthewson 2004, Davis, Gillon, and Matthewson 2014, Bochnak and Matthewson 2020) or from published sources:

(4) Mam varities discussed in this paper:

a. San Juan Atitán (SJA) Mam (author fieldnotes)

b. Todos Santos (TS) Mam (author fieldnotes)

c. San Ildefonso Ixtahuacán (Ixt) Mam (Nora England's publications)

In broad focus declaratives, Mam's word order is VSO (England 1991), although Ā-movement operations (focus, *wh*-questioning, relativization) may move elements to the clausal periphery. A typical broad focus transitive clause is given below, showing V, S, and O constituents. Note that the element labeled 'V' is a complex constituent that may contain a number of elements along with the verb stem, including aspect/mood marking, person and number agreement, and directionals.

(5)

		V			S		O	
[Nti'	ma	tz'=ok	ky-ke'y-an	] [qa	xjal	] [jun	ja	]
[NEG	PROX	B2/3SG=DIR:in	A2/3PL-see-DS	] [PL	person	] [INDF	house	]

'The people didn't see a house.' (SJA Mam; Scott 2023: 56)

Mam is an ergative-absolutive language, with grammatical relations established through verbal agreement. Ergative (**Set A**) cross-references transitive subjects, whereas absolutive (**Set B**) cross-references transitive objects and intransitive subjects.[3] In Mam, absolutive agreement precedes ergative agreement, and, when present, the directional intervenes between absolutive and ergative agreement. We present a transitive verbal template in (6a), and two exemplar sentences showing the realization of this template: in (6b), we see a transitive clause with ergative and absolutive agreement, along with a directional; in (6c), we see an intransitive clause with only absolutive agreement and no directional.

(6) a. Transitive verb template:
NEG/ASP – **Set B** (ABS) – DIR – **Set A** (ERG) – ROOT – SUFFIXES

b. Ma **chin** ok **t-**tzeeq'a-n=a.
PROX B1SG DIR A2/3SG-hit-DS=DA
'You hit me' (Ixt Mam; England 1983a:2)

c. Ma **chin** b'eet=a.
PROX B1SG walk=DA
'I walked' (Ixt Mam; England 1983a:2)

Last to note are the verbal suffixes that occupy the rightmost position within the verb stem (excluding any enclitics that are more rightward). These suffixes include voice morphology and other types of verbal derivation, as well as one of two suffixes that track transitivity. First, all transitive verbs (with just one exception being *il* 'see') surface with the glottalized nasal transitive suffix *-'n* (glossed "DS" for "directional suffix" for the moment; this will be decomposed and modified in section 6); for phonological reasons, the glottal component may surface within the verb stem (see England 1983b: 53-54). Second, the vast majority of unergative intransitives surface with the intransitive nasal suffix *-n* (glossed "AP" for "antipassive"; again, we modify this in section 6), whereas unaccusative intransitives bear no dedicated suffix.

3 Two contributions for directionals in Mam

In this section, we describe Mam directionals in more detail, and make explicit their two contributions. We will ultimately propose that these two contributions—the "lexical" and the "functional"—be disentangled.

Mam features a suite of 12 directionals, presented in the table below. Directionals fit into three major categories, as discussed in Elkins 2023. The first category are the egocentric directionals (e.g., *tzaj* 'come, towards speaker'), which refer to motion with respect to the speaker as the deictic center. The second category are the cardinal directionals (e.g., *jaw* 'up; to the north'), which refer to motion without reference to the speaker at the deictic center. The third and final category contains just the directional *b'aj* 'complete', which does not refer to direction of motion *per se*, but instead to a completed action.

[3] In both Todos Santos and San Juan Atitan Mam, Set B does not cross-reference the features of transitive objects. Instead, Set B on transitive verbs always appears in the default 2/3SG form, and the features of the object are realized in object position, following the subject. The details are not relevant for the present analysis, and we refer the reader to Chapter 3 of Scott (2023) for discussion and analysis of this pattern.

Table 1: Directionals in San Juan Atitán Mam; Scott 2023: 66

Directional	Meaning	Directional	Meaning
xi'	go, away from speaker	*el*	out; to the west
tzaj	come, towards speaker	*ok*	in; to the east
ul	arrive here	*kyaj*	remain
pon	arrive there	*aj*	return
jaw	up; to the north	*iky'*	pass
kub'	down; to the south	*b'aj*	complete

The set of 12 directionals given above are all derived from intransitive verbs of motion, and many have undergone phonological truncation, such as the reduction of a long vowel to a short vowel. The intransitive verbs from which the directionals are derived are still present in the language, as shown in (7). Here, the directional *ul* 'arrive here' is clearly derived from the intransitive verb *uul* 'arrive here'. The verb *uul* in (7) is not functioning as a directional auxiliary, but as an intransitive (directional).

(7) Ma tz=**uul** Noé.
PROX B2/3SG=arrive.here Noé
'Noé arrived here.' (TS Mam)

Directional auxiliaries must combine with a main verb, preceding that main verb and appearing between **Set B** and **Set A** morphology. A directional is often pronounced, along with the relevant **Set B** morphology, as its own prosodic word, as in (8) below.

(8) Ma **tz=ul** **t**-ii-'n Noé kiẍ.
PROX B2/3SG=DIR:arrive.here A2/3SG-bring-DS Noé fish
'Noé brought the fish here.' (TS Mam)

The suite of 12 directionals given in Table 1 may additionally be stacked; that is, two or more directionals may occupy the directional slot in the verb template. When directionals are stacked in this manner (forming 'complex' directionals), there is a predetermined linear order of the stacked elements: a cardinal directional precedes an egocentric one, and an egocentric one precedes *b'aj* (England 1976; Scott 2023: Table 2.26). When stacked, directionals typically undergo further phonological truncation. In example (9a), the directionals *kub'* 'down' and *xi'* 'go, away from speaker' become *ku'x*; in (9b), *jaw* 'up, to the north' and *tzaj* 'come, towards speaker' become *jatz*. We represent complex directionals in our examples as stacked clitics.

(9) a. Ma ∅ **ku'=x** n-awa-'n kjo'n.
PROX B2/3SG DIR:down=DIR:in A1SG-plant-DS corn
'I planted the corn.' (TS Mam)

b. Ma ∅ **ja=tz** n-baq'o-'n jun mata is.
PROX B2/3SG DIR:up=DIR:come A1SG-harvest-DS INDF MW potato
'I harvested some potatoes.' (TS Mam)

Finally, note that when used as main verbs, verbs from which directionals are derived may combine with another directional auxiliary verb. In such cases, the main verb also undergoes truncation:

(10) Ma tz'=**e=pn**=i.
PROX A2/3SG=DIR:out=DIR:arrive.there=DISAGR
'You arrived out west.' (SJA Mam, Scott 2023: 70)

3.1 The "lexical" contribution of directionals

As mentioned above, we argue in this paper that directionals have two major contributions, which we aim to disentangle and describe separately. We begin with what we refer to as their "lexical" contribution. For some verbs, directionals seem to only add "directional", "locative" or "aspectual" content to the verbal event. For example, certain verbs like *ajqelil* 'run' do not require directionals at all. However, as we see in (11b,11c), the addition of a directional adds a more explicit direction of motion to the verbal event.

(11) a. Ma chn= ajqel-n=i.
PROX B1SG= run-AP=DA
'I ran.' (SJA Mam)

b. Ma chn= **el** ajqel-n=i.
PROX B1SG= DIR:out run-AP=DA
'I ran out.' (SJA Mam)

c. Ma chn= **ok=x** ajqel-n=i.
PROX B1SG= DIR:in=DIR:go run-AP=DA
'I ran in.' (SJA Mam)

It is important to flag again here that the verb 'run' in (11) does not *require* a directional; that is to say (11a) is a complete, grammatical utterance. It is also the form that is offered in context-neutral translation tasks. Additionally, when a directional is inserted, there is a transparent semantic contribution, namely direction of movement.

3.2 The "functional" contribution of directionals

We saw above that for some verbs, directionals are not required; if they are added, they have a transparent semantic or "lexical" contribution. For other verbs, though, directionals *are required*. For example, the transitive verb *ke'yil* 'see' may not be used unless a directional is also present within the verbal complex. The directional that typically co-occurs with this verb is *ok* 'in,' which has a less transparent semantic contribution; compare this with the straightforward semantic contributions of the directionals in (11b,11c).

(12) Ma tz=***(ok)** q-ke'y-n=i a=y.
PROX B2/3SG=DIR:in A1PL-see-DS=DA DET=DA
'We saw you.' (SJA Mam; Scott 2023: 67)

In contrast to what we have called the “lexical” contribution of directionals, the directional in (12) is required for grammaticality. When used in this way, directionals often do not have a straightforward transparent semantic contribution with regard to the motion of the verbal event. We refer to this as the “functional” contribution of directionals.

The main focus of this paper is to analyze the functional contribution of directionals; that is, what is the purpose of directionals in active sentences like (12), and why are they required? In the next section, we show that certain classes of verbs can be distinguished based on their behavior with respect to directionals. Specifically, only transitive verbs and unaccusative verbs require directionals. In contrast, unergative verbs, as in (11), do not require directionals to form grammatical clauses. Since unaccusative and transitive verbs differ from unergative verbs in involving theme arguments, the natural hypothesis that arises given these data is that directionals in sentences like (12) are needed to introduce themes. The aim of section 5 will then be to account for this fact.

4 Directionals and argument structure

We discuss three predictions that follow if directionals are needed to introduce theme arguments in Mam:

1. Transitive active verbs should require directionals → 99% true; §4.1.
2. Unaccusative active verbs should also require directionals → true; §4.2.1.
3. Unergative active verbs should never require directionals → true; §4.2.2.

Below we demonstrate that all these predictions are borne out. Section 4.1 focuses on transitive verbs, showing that they always require directionals (though we discuss one exception). Then we turn to intransitive verbs, showing that there is an asymmetry with respect to the behavior of unaccusatives and unergatives: whereas unaccusative verbs require directionals (section 4.2.1), unergative verbs never do (section 4.2.2). Again, the empirical generalization that emerges is that directionals, at least with active verbs (we discuss other voices in section 5), are needed whenever a theme is present.

4.1 All (but one) transitive verbs require a directional

Beginning with transitive clauses, it has long been known that directionals are required to co-occur with transitive verbs in Mam. This observation was first made by England (1976), who summarizes the state of affairs as follows.

> *Most transitive verbs in Mam almost always require directionals. While theoretically possible to use these verbs without a directional, the native speaker much prefers to use one, and even has difficulty accepting forms without directionals for most transitive verbs. Therefore in eliciting a finite transitive verb, the form given almost always includes a directional. If the form is requested without a directional, the native speaker invariably responds with an intransitive rather than transitive form.* (p. 204)

We already presented an example of a transitive clause with its co-occurring directional in (12); two more examples are provided below. Here, we see that the transitive verb *laq’ol* ‘buy’ requires a directional, although the precise choice of directional varies depending on the context and sentence.

(13) a. Ma tz'=***(el)** t-laq'o-'n Xwan tx'otx'.
PROX B2/3SG=DIR:out A2/3SG-buy-DS Xwan land
'Xwan bought land.' (SJA Mam; Scott 2023: 76)

b. Ma ∅ ***(tzaj)** t-laq'o-'n Xwan jun xkoy.
PROX B2/3SG DIR:come A2/3SG-buy-DS Xwan one tomato
'Xwan bought a tomato.' (SJA Mam; Scott 2023: 76)

We stress here that while the choice of a directional might sometimes "make sense lexically" (e.g., 'run out' in (11b)), its core lexical content is often opaque when required by the verb, as above. Table 2 provides a sample of transitive verbs; the directionals we indicate in the table are those which seem to be one of the most often used for each verb in elicitation sessions.[4]

Table 2: Sample of transitive verbs in Mam

Directional + Verb	Meaning	Translation
**(ku'x) + awa'n*	down.go + plant	plant
**(tzaj) + laq'o'n*	come + buy	buy
**(ok) + b'yo'n*	in + hit	hit
**(ok) + ke'yan*	in + see	see
**(tzaj) + q'i'n*	come + take	bring
**(jaw) + xk'lo'xan*	up + wrap	wrap
**(kub') + qesa'n*	down + cut	cut
**(xi) + chmo'n*	go + weave	weave
**(xi) + yek'an*	go + show	show
*(*DIR) + il*	(*DIR) + see	see

As foreshadowed above, the only clear exception to the principle that Mam transitives require directionals is *il* 'see.' In fact, as indicated in (14), this verb seems to *never* allow a directional (see also Scott 2023: 69). We will return to this exception below in section 5; for now, however, it is important to note that there is another verb meaning 'see', *che'yil* (*ke'yl* in SJA) in (15), which is not exceptional in this way. In other words, the exception is specifically with the root *il*, whose lexical idiosyncrasy happens to disallow directionals.

(14) Ma ∅ **{*xi',*tzaj,*kub'...}** w-il xuuj.
PROX B2/3SG DIR:go;DIR:come;DIR:down A1SG-see woman
'I saw the woman.' (TS Mam)

(15) Ma tz'=***(ok)** n-che'y-n=i
PROX B2/3SG-DIR:out A1S-see-DS=DA
'I saw you.' (TS Mam)

[4]In Table 2, some of the morphology from the verbal complex has been removed, namely, aspect and both Set A and B agreement. However, the verb roots are shown with the suffix -*'n* that we gloss as "DS" for now (though see section 6). Keeping this morphology helps to illustrate that this suffix is found only in transitive contexts.

4.2 Intransitives: unergatives vs. unaccusatives

We now turn to the behavior of intransitive verbs in Mam with respect to their interaction with directionals. As we will show, we find an asymmetry between unergative and unaccusative verbs, in which the former does not require directionals but the latter does.

In terms of our categorization of unergative *vs.* unaccusative verbs, we find that they are clearly semantically and morphologically distinct. Unergatives denote agentive events and end with a nasal suffix *-n*, which is also found on derived antipassive verbs (and as such has historically been glossed "AP" for 'antipassive'):

(16) Ma qo scha-**n**=i.
PROX 1PL play-AP=DA
'We played.' (unergative; SJA Mam)

On the other hand, unaccusative verbs in Mam denote non-agentive, directed motion, or change-of-state events, and do not end with *-n*:[5]

(17) Ma ∅ ***(kub')** pax xk'utz'ib'il.
PROX B2/3SG DIR:down break computer
'The computer broke.' (unaccusative; SJA Mam)

Importantly, these two classes of verbs show differential behavior with respect to directionals. Whereas unaccusatives, or at least those that are underived (we explain what we mean by 'underived' below), require a directional (§4.2.1), unergatives never require one (§4.2.2).

4.2.1 Unaccusative verbs require a directional

Beginning with unaccusative verbs in Mam, we find that they consistently *require* directionals. Table 3 shows a representative sample of these verbs and the directionals that typically co-occur with them.

Table 3: Underived unaccusative verbs

Directional + verb	Meaning	Translation
**(jaw) + tolj*	up + fall	fall (animate)
**(etz) + tz'aq*	out.come + fall	fall (inanimate)
**(kub') + pax*	down + break	break
**(el) + neq't*	out + melt	melt
**(txi) + tzaq*	go + rip	rip

While an example sentence with a basic unaccusative verb was already provided above, we provide a second one in (18).

[5] Note that the inventory of unergative vs. unaccusative verbal roots in Mam resembles those reported for other Mayan languages, including Tsotsil (Haviland 1994), Ch'ol (Gutiérrez Sánchez 2004; Coon 2010), and Chuj (Coon 2019). Moreover, similar morphological facts as in Mam obtain in some of these languages. In Chuj, for example, unergatives are derived with overt morphonology *-w* (AP), but not unaccusatives (Coon 2019).

(18) Ma tz'=***(e=tz)** tz'aq k'ab'il t-wi' mes.
PROX B2/3SG=DIR:out=DIR:come fall cup A2/3S-RN:top table
'The cup fell off the table.' (unaccusative; SJA Mam)

We note that it is important to distinguish the behavior of "underived unaccusatives" (as above) from unaccusatives derived via passive morphology. As shown in (19a), directionals are actually *optional* with unaccusative verbs like *b'uch-j* 'shattered', which are derived from a passive suffix (here *-j*) (compare the active use of this root in (19b), in which the directional is obligatory). These data are treated with more depth in section 5.4.2. In short, in that section we will suggest that directionals are no longer obligatory in such cases, because passive morphemes are also tied to the introduction of themes. We additionally find that without derivation, roots like *b'uch* are only possible as stative (non-verbal) predicates (19c).

(19) Unaccusative verb derived with passive morphology

a. Ma ∅ **(kub')** b'uch-**j** vas.
PROX B2/3SG DIR:down shatter-PASS glass
'The glass shattered.' (passive; SJA Mam)

b. Ma ∅ ***(kub')** b'u'ch-an Pey vas.
PROX B2/3SG DIR:down shatter-DS Pey glass
'Pey shattered the glass.' (transitive; SJA Mam)

c. B'u'ch vas.
shattered glass
'The glass is shattered.' (non-verbal predicate; SJA Mam)

Lastly, note that when used as main verbs, the verbs that directionals derive from never seem to require *another* directional. This is shown below. (See Table 1 for full list of directional verbs, and (10) for evidence that such verbs are nevertheless not incompatible with a directional.)

(20) Ma ∅ pon=i
PROX B2/3SG arrive.there=DA
'You arrived there.' (directional as main verb; SJA Mam

4.2.2 Unergative verbs never require a directional

The behavior of unaccusatives contrasts with what is observed for unergative verbs, which *never* require a directional (see also (11a) and (16) above). A sample of unergative verbs is provided in Table 4. Recall that unergative verbs are universally derived with the antipassive suffix *-n*.

(21) Ma qo xnaq'tza-n.
PROX B1PL listen-AP
'We studied.' (unergative; SJA Mam)

[6] In Table 4, some of the morphology from the verbal complex has been removed, namely, aspect and Set B agreement. However, the verb roots are shown with the suffix *-(a)n* that we gloss as "AP" for now. Keeping this morphology helps to illustrate that this suffix is found only in agentive contexts.

[7] For a more nuanced analysis of the meanings of the different verbs for eating and drinking, we refer the reader to England 1980.

Table 4: Sample of unergative verbs[6]

Verb	Translation	Directional	Translation
yon	wait	*sch'in*	read/shout
ajqelan	run	*tz'ib'an*	write
b'ixan	dance	*schan*	play
b'itzan	sing	*aq'nan*	work
yolan	talk	*xnaq'tzan*	study
b'in	listen	*wan*	eat food
tan	sleep	*lon*	eat fruit
na'n	pray	*kxun*	eat sth. crunchy[7]
lipan	fly	*chyon*	eat meat
b'et(an)	walk	*k'an*	drink

In translation tasks with no previous situational context provided to consultants, unergative verbs are rarely if ever offered with a directional; and if one is provided, omitting the directional always results in an acceptable sentence. We propose that this optionality falls out from the *lexical contribution* of directionals: unergatives can appear with one whenever it is relevant to state information about the motion of the event, as in (22). In such cases, the deictic contribution tends to be much more transparent.

(22) Ma chn= **el** ajqel-n=i.
PROX B1SG= DIR:out run-AP=DA
'I ran out.' (unergative; SJA Mam)

4.3 Summary

Table 5 summarizes the empirical observations described in this section. All transitive (with one known exception) and (underived) unaccusative verbs must co-occur with directionals. Unergatives, on the other hand, never require one.

Table 5: Directionals and argument structure

	Arguments		DIR required?
Transitive	Agent	**Theme**	99%
Unaccusative		**Theme**	yes
Unergative	Agent		no

As also indicated in the table, the necessity of a directional correlates almost perfectly with the presence of an internal argument (at least with active verb forms). That is, while transitives and unaccusatives both take internal arguments, unergatives do not.

In the next section, we propose that we can make theoretical sense of these empirical generalizations if Mam directionals are needed to introduce themes, explaining their functional contribution. In other words, Mam verbs in general do not introduce *any* arguments: they are predicates of events (as in Castañeda 1967, Borer 2005, Pietroski 2005, Levinson 2010, Williams 2009, Londahl 2014, Elliott 2020). Directionals are the only exception: they introduce internal arguments, and are thus needed to express basic transitive and unaccusative clauses.

5 Severing the *internal* argument from Mam verbs

This section is structured as follows. Section 5.1 first provides the theoretical background on which our analysis of Mam argument structure is built. Sections 5.2 and 5.3 then provide our analysis. Finally, section 5.4 discusses voices other than the active voice and their effects on directional syntax. As we will see, in both antipassive and passive voices, directionals are no longer obligatory, which we interpret as welcome implications of our proposal.

5.1 Assumptions about argument structure and the syntax of the extended VP

Schein (1993) and Kratzer (1996), building on previous work on argument structure (Williams 1981, Marantz 1984, Hale and Keyser 1993), influentially proposed that external arguments should be severed from the lexical semantics of verbal roots in languages like English (see also works like Pylkkänen 2008 and Williams 2015, as well as references therein). Under this analysis, instead of taking both an external and internal argument, as in other related approaches (such as some branches of Davidsonian (Davidson 1967) or neo-Davidsonian semantics), transitive verbs only take an internal argument. A classic entry for the transitive verb 'to feed' under this approach, which we will be assuming here, is provided in (23a). Kratzer (1996) proposes (23b) instead.

(23) a. $[\![\textit{feed}]\!] = \lambda x.\lambda y.\lambda e.\ \textsc{feed}(x)(y)(e)$
b. $[\![\textit{feed}]\!] = \lambda x.\lambda e.\ \textsc{feed}(x)(e)$

Under this approach, transitive and unaccusative roots have the same basic argument structure. Compare, for instance, the transitive root in (23b) with the unaccusative root in (24).

(24) $[\![\textit{fall}]\!] = \lambda x.\lambda e.\ \textsc{fall}(x)(e)$

How does the external argument enter the derivation in a theory like the one above, where transitive roots, like the one in (23b), do not encode an extra argument corresponding to it? These works propose that they be delegated to a functional head within the verbal domain, which we will represent as "v/Voice$_{\text{AGT}}$" here following Coon 2019 (this head is sometimes alternatively labeled as v or Voice; we will motivate the use of v/Voice below in section 5.3):

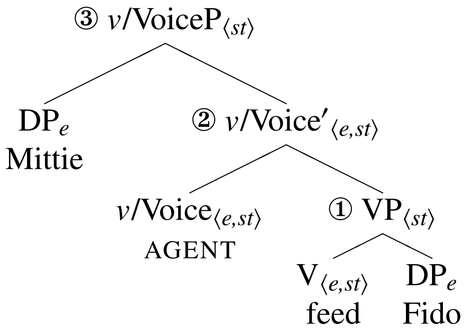
(25) $[\![v/\text{Voice}_{\text{AGT}}]\!] = \lambda y.\lambda e.\ \textsc{agent}(e) = y$

This functional head merges with the VP (itself a function from events to truth values, i.e. type ⟨s,t⟩), via a compositional rule called "Event Identification", a variant of Predicate Modification insofar as it entails the conjunction of two functions (Heim and Kratzer 1998). This is to allow type $\langle e,st \rangle$ functions to combine with type $\langle s,t \rangle$ functions. Throughout this paper, we will be assuming existential closure of the event argument higher in the structure.

To illustrate how the composition of the extended VP unfolds, consider the example in (26). In step ①, the V and the theme, in this case Fido, first compose via Function Application (FA)

(Heim and Kratzer 1998) to deliver a predicate true of events of feeding Fido. Second, Event Identification in step ② allows for the VP and *v*/Voice to compose, creating a type ⟨*e,st*⟩ function that can now take an agent argument. The DP agent, here Mittie, is then introduced via FA in step ③.

(26) Mittie fed Fido



③ $[\![v\text{/VoiceP}]\!]$: $\lambda e.\ \textsc{feed}(\text{F})(e)\ \&\ \textsc{Agent}(e) = \text{Mittie}$

② $[\![v\text{/Voice}']\!]$: $\lambda x.\lambda e.\ \textsc{feed}(\text{F})(e)\ \&\ \textsc{agent}(e) = x$

① $[\![\text{VP}]\!]$: $\lambda e.\ \textsc{feed}(\text{Fido})(e)$

With this theoretical background in place, we now turn to Mam verbs. We have two aims. First, we must understand how Mam verbal roots and directional roots are lexically specified. In particular, given the hypotheses put forth in Section 4, the entry for regular verbal roots in Mam should differ from the one in (23b), insofar as internal argument should be dissociated from lexical verbs and instead incorporated into the argument structure of directional roots. Second, we must provide a syntactic composition for the extended verbal domain that integrates directionals. The next two subsections address these two aims, respectively.

5.2 Mam verbal roots and argument structure

This section is divided into two parts. We first discuss regular verbal roots in section 5.2.1, turning to directional roots in section 5.2.2.

5.2.1 Verbal roots as properties of events

As we saw in section 4, the vast majority of Mam verbs (with one exception) cannot themselves introduce internal arguments. As foreshadowed above, we propose here that this is because internal arguments in Mam are also severed from verbal roots, which merely denote properties of events.[8] Consider, for instance, the following entries for transitive and unaccusative verbal roots, which we essentially take to be argumentless (apart from the event argument):

(27) a. $[\![\textit{feed}]\!] = \lambda e.\ \textsc{feed}(e)$ (Mam transitive verb)

b. $[\![\textit{fall}]\!] = \lambda e.\ \textsc{fall}(e)$ (Mam unaccusative verb)

[8] It is important to note that other work has also proposed this, even for English. See for instance Castañeda 1967, Borer 2005, Pietroski 2005, Williams 2009, Levinson 2010, Londahl 2014, Elliott 2020, and many more. Here, we maintain the analysis by Kratzer (1996) for English, as it allows us to make sense of the difference between English and Mam, and Mayan-internally, Mam and Chuj (see Coon 2019). Allowing variation in whether internal arguments may be inherently selected by verbs also allows us to handle Mam-internal variation, and in particular, exceptional verbs like *il* 'see', which never combine with a directional (see data in (40) and (41) and the paragraph above it). To be clear, however, a null functional head could also be responsible for introducing the internal argument, even in languages like English and Chuj.

The entries in (27) differ in a crucial way from those proposed for English by Kratzer (1996). While this work encodes some argument structure within verbal roots (the internal argument), the roots in (27) involve *no* argument structure: they merely denote properties of events (compare (27a) with (23b)). We propose instead that the introduction of internal arguments in Mam is delegated to directionals (see also section 5.4.2 on passives), which we turn to in the next subsection.

Before moving on, however, one last point about root classes is in order. Notice in (27) that the entries for transitive and unaccusative verbs are not distinguished in terms of their argument structure: both denote properties of events. This aligns with proposals such as Kratzer 1996 or Davis 1997 for languages like English or St'át'imcets (see (23b) and (24)). It also aligns with a proposal by Coon (2019) for the Mayan language Chuj. Indeed, Coon proposes a Krazterian semantics for Chuj roots, where transitive and unaccusative roots are not distinguished in terms of their argument structure (see also Coon and Royer 2020). Now, while it is not necessary to classify Mayan roots in terms of their argument structure, Coon (2019) shows that roots are not completely acategorical (see also Haviland 1994, Henderson 2019). For verbal roots, transitive (or 'agentive') roots can be distinguished from unaccusative (or 'non-agentive') roots with regards to their stem-forming possibilities. This means that the two classes of verbal roots directly dictate the voice and valence functional morphology that may combine with them. In this work, we build on Coon (2019) (citing Levin and Rappaport-Hovav 1995) in assuming that the class of the root is determined based on whether or not the root is semantically compatible with external agency. In brief, we assume that roots like 'feed' in (27a) imply external agency (feeding must be done by someone), and thus an agent argument will have to reflected in the syntax (such as via an agentive functional head), whereas verbs like 'fall' in (27b) imply a lack of external agency, and are not compatible with agentive functional heads.

5.2.2 Directional auxiliary verbs as theme introducers

We now turn to the semantics of directional auxiliary verbs. We saw in section 4 that whenever an active verb combines with a theme (again with one exception), a directional is needed. To account for this, we propose that the introduction of internal arguments in Mam is delegated to the closed class of directional auxiliary verbs described in section 3. In particular, we propose the following denotation for directionals:[9]

(28) $[\![\textsc{dir}]\!] = \lambda x.\lambda e.\ \mathbf{dir}(e)\ \&\ \textsc{theme}(e) = x$ (Mam directional)

Under this account, directionals are elements of the verbal domain that serve (i) to specify information about the direction, time, or location in which an event took place (abbreviated as "DIR(e)" in (28)), and (ii) to introduce the internal argument ("THEME(e)" in (28)). The entry in (28) can account for two core facts about directionals. First, it explains why directionals, when used as main verbs of motion, can exceptionally combine with a theme without any auxiliary functional item:

(29) Ma tz=**uul** Noé.
PROX B2/3SG=arrive.here Noé
'Noé arrived here.' (TS Mam, repeated from (29))

[9]Note that we include theme in the metalanguage for convenience. Crucially, though, the directional introduces the theme, i.e., the use of "theme" should not be interpreted as indicating its separation from the directional.

Second, the entry in (28) can account for the fact that directionals get employed as auxiliary verbs in the language, whenever a theme must be introduced. In other words, it explains the "functional contribution" we described in section 3.2. Importantly, we assume that once a theme argument is introduced by a head such as Dir in the verbal spine, the internal argument position is saturated and cannot be introduced again higher in the structure.

At this point, we must address the fact that Mam directionals cannot *categorically* involve argument introduction. Indeed, as discussed in section 3.1, they are sometimes employed in positions in which they arguably do not introduce any argument. We called this the "lexical contribution", since in such cases directionals only add information about the motion of the event described by the verbal complex. The examples below highlight the lexical use again. In (30), the verb in question is unergative, and so the directional can clearly not be introducing a theme. In (31), we see that directionals may undergo stacking. Whenever stacking occurs, it must be the case that at least one of the two directionals does not involve argument introduction (regardless of verb class).

(30) Ma chn= **el** ajqel-n=i.
PROX B1SG= DIR:out run-AP=DA
'I ran out.' (SJA Mam; repeated from (11b))

(31) Ma ∅ **ja=tz** n-baq'o-'n jun mata is.
PROX B2/3SG DIR:up=DIR:come A1SG-harvest-DS INDF MW potato
'I harvested some potatoes.' (TS Mam, repeated from (9b))

For now, we will therefore assume a systematic lexical ambiguity corresponding to directional items in Mam. While directionals often introduce an argument (28), they sometimes do not:[10]

(32) $[\![\text{DIR}_2]\!] = \lambda e.\ \mathbf{dir}(e)$ (second entry for Mam directionals)

In the rest of this paper, we will focus on cases where directionals do introduce internal arguments, as in (28). However, we highlight that the systematic ambiguity proposed above seems motivated. Across Mayan languages, directional auxiliary verbs have grammaticalized from a closed class of unaccusative verbal roots that freely combine with internal arguments. For all Mayan languages we know of, the relevant roots remain (for the most part) synchronically active. The entries in (28) and (32) thus aim to capture the double function of directionals, an apparently widespread characteristic of Mayan languages (see e.g., Mateo Toledo 2023).

Having provided lexical entries for verbal roots and directional auxiliary verbs (in particular the one in (28)), we turn in the next subsection to the external syntax of directionals, and how the composition of the Mam verbal domain unfolds.

5.3 Composing the extended VP domain in Mam

Here, we schematize how the VP domain in Mam is composed, paying special attention to the functional contribution of directionals. We provide an example of a transitive derivation, using

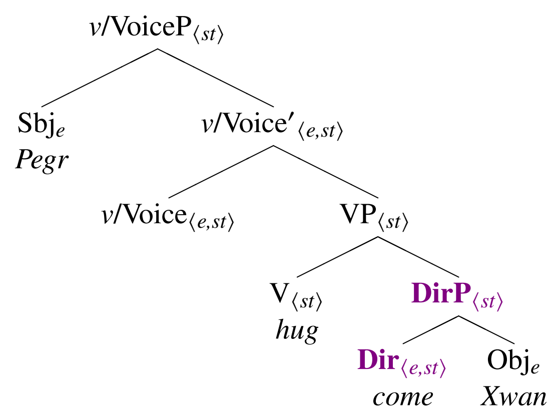
[10] Another way to account for the purely lexical use without resorting to an ambiguity could be to posit that directionals can spellout syntactic spans or chunks of structure involving DirP and, optionally, an FP that introduces the theme (see e.g., Starke 2009; Svenonius 2016). We also saw that when used functionally, the semantic contribution of directionals is sometimes bleached. We hypothesize that directional auxiliary verbs might in fact be undergoing a stage of grammaticalization, in which their directional meaning sometimes gets lost (in a way reminiscent to phrasal verbs in English). We leave an exploration of these different topics for future work.

(33) as an example.

(33) Ma ∅ ***(tzaj)** t-xhq'e-'n Pegr Xwan.
PROX B2/3SG DIR:come A2/3SG-hug-DS Pegr Xwan
'Pedro hugged Juan.' (TS Mam)

Building on Elkins (2023), we propose the minimal syntax in (34) for the extended VP domain (ignoring tense-aspect morphology). Here, the directional projection (DirP) is complement to V; it is the directional itself which selects for the internal argument of the verb.[11] We also make two additional assumptions regarding clause structure, which both build off previous literature on the syntax and semantics of Mayan languages. First, verb-initiality in Mam is derived via head-raising of the verb to a position at least above *v*/VoiceP shell (often labeled "SSP" in the Mayanist literature, following Clemens and Coon 2018 and Elkins 2023). For Mam specifically, we propose that directionals are also part of this head-movement chain, deriving the mirror DIR–VERB–VOICE morphological order (see section 6 for a concrete implementation). Second, we assume consistent object raising in Mam (see Coon, Mateo Pedro, and Preminger 2014, Coon, Baier, and Levin 2021, Royer and Coon 2025; as well as Scott 2023 and Myers 2025a specifically on Mam), which we ignore in the trees for space. Object raising is independently needed to account for patterns of syntactic ergativity and binding (Coon et al. 2014, Coon et al. 2021, Royer 2025, Scott 2023), as well as split ergativity in non-finite clauses (Elkins, Royer, and Scott 2025). We crucially assume, however, that the object is interpreted in its base position for purposes of semantic interpretation.

(34) Tree for (33) (*Pedro hugged Xwan*)



Finally, note that we build on Coon and Preminger 2013 and Coon 2019 on other Mayan languages in assuming that the agent is introduced in a bundled *v*/Voice projection (see also Pylkkänen 2002, 2008, Harley 2017; see Ranero 2021 and Burukina and Polinsky 2025 for a different approach in Kaqchikel). As will be discussed in the sections to come, this projection encodes information typically associated with both *v* and Voice (Alexiadou, Anagnostopoulou, and Schäfer 2015). Like *v*, it combines with a root and encodes categorizing morphology. Like Voice, it is involved in the

[11] Another possible option for the structural position of the directional would be for it to attach between VP and *v*/VoiceP (or even in a higher functional projection above the *v*/VoiceP), introducing the theme in its specifier. It would then compose with the rest of the verbal spine through Event Identification. In section 6, we provide one reason to adopt the lower structure in (34): once head movement is assumed (following Clemens and Coon 2018, Elkins 2023), this merge order derives the correct linear order of the elements of the verbal complex.

introduction of external arguments and is sometimes responsible for case assignment (see Coon 2019, Deal and Royer 2025).

With this syntax in place, consider now the entries in (35), which assume the theory proposed in section 5.2. (Note that in section 6, we propose that agentive *v*/Voice in (35c) is in fact overtly instantiated in Mam, via the suffix *-n*. As this section aims to focus on directionals, we have chosen to not represent this in the tree in (34). However, given the theory proposed in section 6, note that the suffix *-n* would appear under *v*/Voice in (34).)

(35) a. $[\![\textit{xhq'e}_{\text{HUG}}]\!] = \lambda e.\ \text{HUG}(e)$

b. $[\![\textit{tzaj}_{\text{COME}}]\!] = \lambda x.\ \lambda e.\ \text{THEME}(e) = x\ \&\ \text{TOWARD.SPEAKER}(e)$[12]

c. $[\![v/\text{Voice}]\!] = \lambda x.\ \lambda e.\ \text{AGENT}(e) = x$

In (35a), the verb denotes simply an event. The directional therefore has two roles: the first is to indicate information about the directionality of the event, and the second is to introduce the theme. Finally, the entry *v*/Voice in (35c) will allow for the introduction of the agent via Event Identification (Kratzer 1996).

For illustration, a bottom-up derivation for the sentence in (34) is provided in (36-39).

(36) $[\![\textbf{DirP}]\!]$: $\lambda e.\ \text{TWRD.SPKR}(e)\ \&\ \mathbf{Th(}e\mathbf{) = Xwan}$

[**Dir** *tzaj*] [Obj *Xwan*]

(through Function Application)

(37) $[\![\text{VP}]\!]$: $\lambda e.\ \mathbf{Hug(}e\mathbf{)}\ \&\ \text{TWRD.SPKR}(e)\ \&\ \text{TH}(e) = \text{Xwan}$

[V *xhq'e*] [[**Dir** *tzaj*] [Obj *Xwan*]]

(through Predicate Modification)

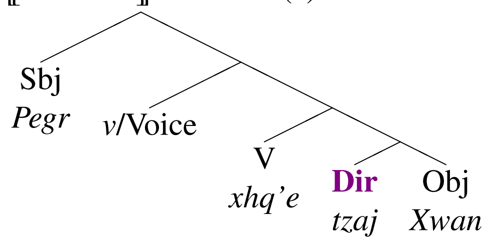
(38) $[\![v/\text{Voice'}]\!]$: $\lambda y.\lambda e.\ \text{HUG}(e)\ \&\ \text{TWRD.SPKR}(e)\ \&\ \text{TH}(e) = \text{Xwan}\ \&\ \mathbf{Ag(}e\mathbf{) = }y$

[*v*/Voice] [[V *xhq'e*] [[**Dir** *tzaj*] [Obj *Xwan*]]]

(through Event Identification)

[12] Directionals like *tzaj*, which we describe in §3 above as being speaker-oriented, can nonetheless be used when there are two third-person arguments, like the sentence currently under discussion (33). In such instances, the movement should not be understood as necessarily being undertaken by the theme argument; instead, the verbal event should be understood as happening in a direction toward the utterer of the sentence. In this paper, we do not use a narrower definition of 'theme' which describes the undergoer of movement (Gruber 1965; see also Williams 2015:152-158).

(39) ⟦*v*/VoiceP⟧: λ*e*. HUG(*e*) & TWRD.SPKR(*e*) & TH(*e*) = Xwan & **Ag(*e*) = Pegr**



(through Function Application)

Crucially, this theory of Mam argument structure proposes that internal arguments be severed from verbal roots in Mam. Whereas severing the external argument from the denotation of the verb is familiar from Kratzerian semantics, our novel contribution here is to argue that in Mam, *internal* arguments must also be severed (as also proposed in other works on English, e.g., Borer 2005, Pietroski 2005, Levinson 2010, and Elliott 2020; see also discussion in Williams 2015).

As a final note, this analysis allows us to capture the lexical exception mentioned above. Recall that the one transitive verb that does not require a directional—and in fact cannot occur with one—is *il* 'see' (the relevant example is repeated as (40) below). We propose that this verb is unique in that it exceptionally requires a lexical entry in which the internal argument is already included (41). Being thus specified, the use of a directional is not necessary.

(40) Ma ∅ **{*xi',*tzaj,*kub'...}** w-il xuuj.
PROX B2/3SG DIR:go;DIR:come;DIR:down A1SG-see woman
'I saw the woman.' (TS Mam)

(41) ⟦il_{see}⟧ = λ*x*.λ*e*. SEE(*e*) & THEME(*e*) = *x*

Why only this one transitive verb—and in fact only this verbal root as a whole—is incompatible with directionals in Mam is unclear. However, we hypothesize that the incompatibility is due to the fact that the root *il* itself encodes directional meaning. For one, while this verb is often translated as 'see', one speaker comments that it means something closer to 'perceive'. If this verb uniquely encodes specific directionality in the meaning of the root (e.g., 'perceive' ≈ 'see in'), than perhaps it is incompatible with directional auxiliaries. Moreover, it is important to note again that Mam possesses another verb translating as 'see' (in the more general sense), which does require a directional (see (12) above).

5.4 Voice and its effects on directional syntax

The theory of argument structure proposed above focused on *active* verbs. Here, we turn to the effects of voice (and valency) alternations for directional syntax. Specifically, if our proposal that directionals introduce internal arguments is on the right track, then it makes interesting predictions about the presence/optionality of directionals given different voices. In particular, we discuss two predictions, which—all else being equal—should follow from our analysis:

(42) a. In voices in which themes are absent, such as antipassives, directionals should no longer be needed. (borne out)
b. In voices in which themes are present, such as passives, directionals should in principle still be needed. (not borne out)

In the next subsections, we show that while prediction (42a) is borne out (§5.4.1), prediction (42b) is not (§5.4.2). In other words, directionals are optional not only in the antipassive voice, as expected, but also in a subset of the passive voices found in the language. While this may seem unexpected given mainstream approaches to passives, we do not view the optionality of directionals in passives as a critical argument against our proposal. Instead, we believe that Mam's rich morphology, and in particular its remarkable explicitness in reflecting the introduction of arguments overtly, opens the door to a new theory of passive constructions in Mam, one in which the passive morphology can sometimes reflect an alternative mode of introducing themes.

Before moving on, it is important to highlight that the two predictions above only follow if we consider the "functional contribution" (§3.2) of directionals, that of introducing themes. Since directionals can also have a more "lexical contribution" (§3.1), where they do not introduce any argument, they are predicted to be optional with both voices, as is indeed the case.

5.4.1 Antipassives circumvent the need for a directional

In Mam antipassives, a directional is not required. This distributional fact follows from the idea that directionals introduce theme arguments, while antipassive constructions do not involve a syntactically projected theme. In this section, we first illustrate how antipassives work descriptively in Mam and then provide an analysis in which they are formally indistinguishable from unergatives, following Burukina and Polinsky (2025).

In Mam antipassives, the object is typically either absent (43a) or appears within an oblique (43b), in which case it appears as part of a relational noun phrase. In such cases, a directional is never required, and such constructions in fact typically lack them. A representative active transitive sentence with an obligatory directional is provided in (44) for comparison. Notice that the same verbal root, *b'yo* 'hit', is used in (43b) and (44), suggesting again that it is truly the syntactic presence of a theme that forces the presence of a directional. This is also reflected in the transitivity suffix used; in (43b), the unergative/antipassive suffix *-n* is used, whereas in (44), the glottalized transitive suffix *-'n* is used.

(43) a. Ma chin chmo-n-i.
PROX B1S weave-AP=DA
'I did some weaving.' (antipassive; SJA Mam)

b. N=chin b'yo-n=i t-i=y ew.
IPFV=B1SG hit-AP=DA A2/3SG-RN:PAT=DA yesterday
'I was hitting you yesterday.' (antipassive; SJA Mam; Scott 2023: 80)

(44) N=∅=*(**o'k**) n-b'yo-'n=i a=y ew.
IPFV=B2/3SG=DIR:in A1SG-hit-DS=DA DET=DA yesterday
'I was hitting you yesterday.' (transitive; SJA Mam; Scott 2023: 80)

Mam antipassives are also often employed to facilitate A-bar extraction of the transitive subject (see e.g., Coon, Mateo Pedro, and Preminger 2014, Aissen 2017 and Royer and Coon 2025 on "Agent Focus"), as shown in the contrast in (45a). Again, directionals are typically absent from such antipassivized constructions, thereby contrasting with their active counterparts (45b).

(45) a. Ja xin xjaal e-∅-b'iyoo-**n** [$_{\text{OBL}}$ t-e jel b'alam]
DEM CLF man COM-B2/3SG-kill-AP A2/3SG-RN CLF jaguar
'It was the man who hit the jaguar' (antipassive; TS Mam)

b. E ∅ *(**kub'**) t-b'iyo-'n xin xjaal jel b'alam
COM B2/3SG DIR A2/3SG-kill-DS CLF man CLF jaguar
'The man killed the jaguar' (transitive; TS Mam)

Importantly, these facts naturally follow from the analysis of argument structure proposed in section 5. If the obligatoriness of directionals is truly tied to internal argument introduction, as we proposed, then removing (or demoting) the internal argument should bleed the necessity for a directional. The above examples show that this is the case.

At this point, it is important to be more specific about what is intended by the term "antipassive." As discussed in detail in recent work by Burukina and Polinsky (2025) on the Mayan language Kaqchikel, one common approach to antipassives is to treat them as counterparts of passives: the idea is that since passives tend to entail the existence of an agent or causer (see next subsection), antipassives should also entail the existence of a theme (Polinsky 2017, Basilico 2019, Heaton 2020). Following Burukina and Polinsky (2025), who build on previous work both in Mayan (Coon 2019) and beyond (Aldridge 2012), we believe that this is likely not the right approach of antipassives in Mam. Indeed, Burukina and Polinsky (2025) show that antipassives, realized with the cognate suffix *-Vn* in Kaqchikel, fail all diagnostics suggestive that implicit (grammatically active) themes are present in antipassive constructions. While we leave a full investigation of antipassive voice to future work, it seems to us that this is also right in Mam.

What is more, in section 6, we will show that antipassive and unergative verbal stems take the same suffix, namely *-n* (also true in Kaqchikel; see Burukina and Polinsky 2025: 17). Building on Burukina and Polinsky (2025), we will therefore relate this suffix to agent introduction (see the entry in (53) below) and simply assume, like Burukina and Polinsky, that no implicit theme argument is ever projected in what has been traditionally called the "antipassive" in literature on Mam. Before doing that, however, we turn to a second voice construction and its effects on directional syntax in Mam, namely passives.

5.4.2 Passives also circumvent the need for a directional

In Mam passive constructions, a directional is not required. Given that passive constructions do involve theme arguments as subjects, we propose that passive constructions introduce the theme at passive *v*/Voice itself. While not conventional, our analysis that passives introduce themes in Mam explains the "circumvention" effect: because passive *v*/Voice introduces themes, it is incompatible with a predicate having a functional DIR head with a saturated theme argument. In this section, we illustrate the descriptive properties of Mam passives and argue that they instantiate an alternative locus of theme introduction, distinct from the one found in active transitives and unaccusatives.

In Mam passive constructions, the theme (or patient) of the action is the syntactic subject, coindexed with Set B marking, and the verb bears passive morphology. Mam has at least four different flavours of passives, all of which allow an optional oblique argument.[13] As laid out in Table 6

[13]We note that there is considerable variation in the reported inventory and meaning of passive morphology for different dialects of Mam. See, for instance, Pérez Vail 2014: chapter 3, section 3.1, for an overview. Since more descriptive work is needed on passives, this section only aims to provide a broad overview of the distribution of

(England 1983b, England 2017), the different passives differ with regards to whether the optional oblique argument is interpreted as an agent or a causer, where “agent” describes an argument that voluntarily initiates an action, while “causer” describes an argument that involuntarily causes one. As indicated in the table, the different morphemes also impose different φ-feature restrictions on the optional oblique argument.

Table 6: Passives in Mam (England 1983b)

Morpheme	Oblique Agent	Oblique θ-Role	Oblique φ-feat.	Directional required
-Vt	optional	agent	any	✗
-b’aj	optional	agent	3rd only	✔
-j	optional	causer	any	✗
-njtz	optional	causer	3rd only	✗

The form *-Vt*, identified as *-eet* in England 1983b, 1988, is generally described as the “regular syntactic passive” (England 1988: 534), as it is the most productive of all four forms and occurs with almost all transitive verb roots.[14] A passive-active contrast with this morpheme is provided below. As indicated in the table, with *-Vt* passives, the optional oblique argument is interpreted as the causer of the event:

(46) a. E ∅ pj-**eet** q’a Miguel [$_{\text{OBL}}$ t-u’n xin xjaal]
COM B2/3SG hit-PASS CLF Miguel A2/3SG-by CLF man
‘Miguel was hit by the man’ (passive; TS Mam)

b. E tz’=**ok** t-pju-’n xin xjaal q’a Miguel
COM B2/3SG=DIR A2/3SG-hit-DS CLF man CLF Miguel
‘The man hit Miguel’ (active; TS Mam)

As also indicated in the table—and crucially for the purposes of this paper—directionals are never needed when verbs are derived with three of the four passive morphemes. In fact, in discourse neutral elicitation contexts, these passives are typically never provided with one. The three passive morphemes include the more productive agentive passive provided in (46a), but also the two “causer” passives *-j* and *-njtz*. Even when the same roots generally require a directional when used in the active voice, they are typically not provided with one in the passive. In other words, while a directional is optional in the passive (46a), it is obligatory in the active (46b).

Examples of the *-j* (47a) and *-njtz* (47b) causer passives are provided below for illustration, with translations as indicated in England 1983b: 203-205 (see also England 1988). As England (1983b: 203) notes, the *-njtz* and *j* passives “have the lexical function of indicating that the [transitive subject] has lost, or does not have, control over the action”. Crucially, again, no directional is needed with such passives.

(47) a. Ma ∅ juus-**j** chib’j [$_{\text{OBL}}$ t-u’n Mal]
PROX B2/3SG burn-PASS meat/food A2/3SG-by Maria
‘The food was burnt by Maria.’ (by accident) (Ixt Mam, England 1983b: 205)

different forms of passives, specifically as described in England 1983b, 1988.

[14] As far as we can tell, the four passives below are “designated passives,” meaning that they are not syncretic with other voices (Alexiadou et al. 2015, Oikonomou and Alexiadou 2022).

b. Ma ∅ tzeeq'a-**njtz** Cheep [$_{\text{OBL}}$ t-u'n Kyel]
PROX B2/3SG hit-PASS José A2/3SG-by Miguel
'José was hit by Miguel.' (by accident) (Ixt Mam, England 1983b: 203)

Finally, the last passive morpheme in Table 6 is *-b'aj*. Unlike the three other passive forms, *-b'aj* "requires the use of a directional and includes motion or process in its meaning ('X happened because someone went to do it')" (England 1983b: 206-207):

(48) Ma ch-**ex** q'i-**b'aj** eky' [$_{\text{OBL}}$ t-u'n Mal]
PROX B2/3PL-DIR:go take/bring-PASS hen A2/3SG-by Maria
'María went to bring the hens.' (Ixt Mam, England 1983b: 207)

Against these empirical facts, we can draw two conclusions. First, the need for a directional with *b'aj* follows naturally from our analysis: the root *qi* 'take/bring' does not introduce a theme, and so a directional is needed to introduce the subject in (48). Second, the absence of a directional for all three other passive forms (46a, 47a, 47b) requires an explanation. All else being equal, if (i) verbs in Mam are mere properties of events, introducing neither themes nor agents, and (ii) directionals must normally be employed in order to introduce themes, then this requirement should hold across the board, regardless of voice morphology. While we leave an in-depth analysis of the syntax and semantics of passive constructions for future work, we conclude this section with what we consider to be a plausible explanation for the latter of the two conclusions.

Since a subset of passive morphemes in Mam override the need for a directional, we hypothesize that these morphemes can sometimes "do the work" of the directional, insofar as three of the four passive morphemes in Mam can also introduce theme arguments. Once we view some instances of passive morphology in Mam as an alternative means to introduce themes, then it follows that directionals are no longer necessary. Concretely, we sketch the following entries for each of the relevant passive morphemes, where they instantiate flavors of *v*/Voice heads.

(49) Flavors of v/Voice$_{\text{PASS}}$[15]

a. $[\![\textit{-Vt}]\!] = \lambda x.\lambda e\text{: } \exists y[y = \text{AGENT}(e)].\ \text{THEME}(e) = x$

b. $[\![\textit{-j}]\!] = \lambda x.\lambda e\text{: } \exists y[y = \text{CAUSER}(e)].\ \text{THEME}(e) = x$

c. $[\![\textit{-njtz}]\!] = \lambda x.\lambda e\text{: } \exists y[y = \text{AGENT}(e)\ \&\ \neg\text{PART}(y)].\ \text{THEME}(e) = x$

d. $[\![\textit{-b'aj}]\!] = \lambda e\text{: } \exists y[y = \text{CAUSER}(e)\ \&\ \neg\text{PART}(y)].\ e$

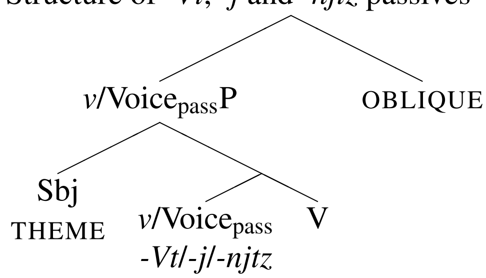
Since the primary distinction in passive constructions involves properties of an implicit or explicit oblique argument, we argue, as standard, that all four passives involve exsistential closure over either an agent or causer (see Coon 2019 and Burukina and Polinsky 2025 for similar approaches to passives in other Mayan languages). However, we further propose that three of the four passive voice morphemes (*-Vt*, *-j*, *-njtz*) are responsible for introducing the theme argument—both syntactically and semantically. Finally, in some cases (49c,d), the passive further constrains the oblique argument's φ-features, such that the oblique argument must be a third person (see England 1983b, 1988, as well as Pérez Vail 2014 for discussion); we represent this for convenience as "¬PART" (to be read as "not a speech act participant").

[15]Note that the passive suffix *-b'aj* is of a different semantic type than the other three passive morphemes. Compositionally, this is not an issue: while the three other passive morphemes will combine with the VP through Event Identification, *-b'aj* will combine with the VP via predicate modification.

Given that most flavors of passive *v*/Voice involve both the agent/causer and theme arguments in their denotation, we may expect that only transitive roots like *laq'ol* 'buy' (which require a semantic agent and are compatible with a theme) are compatible with passive syntax. This is indeed true: unaccusatives and unergatives cannot be passivized. As previously mentioned, we build on Coon (2019) (citing Levin and Rappaport-Hovav 1995) in assuming that root classes are determined based on semantic compatibility with external agency. We thus assume that semantic agent/theme requirements and (in)compatibilities are encoded in the root. For example, (unaccusative) roots like *tz'aq* "fall" are semantically incompatible with agents, and (unergative) roots like *ajqel* "run" are semantically incompatible with themes. Thus, both "intransitive" root types are incompatible with passive syntax in Mam, which includes agents/causers and themes. It follows that passive *v*/Voice can only combine with "transitive" type roots (e.g., *laq'ol* 'buy') that are compatible with both agents and themes.

This unconventional analysis of passives may be justified in Mam, given the atypical behavior of argumentless verbs in the language. Traditional analyses treat passives as promoting themes to subject position via movement of the internal argument (or the "logical object") to subject position (Chomsky 1981, Jaeggli 1986, Collins 2005, a.o.). This view assumes that verbs introduce theme arguments. However, since Mam verbs do not introduce any arguments, the standard passive functional structure does not apply straightforwardly. Instead, we propose that when an argumentless verb combines with one of the three argument-introducing passive *v*/Voice heads in (49a-c), a theme is externally merged in the specifier of this projection. This is schematized in (50):[16]

(50) Structure of *-Vt*, *-j* and *-njtz* passives



In positing that passive morphemes are a flavor of *v*/Voice, it is important to observe that passive morphemes are in parallel distribution with the suffix *-n*. In the next section, we will argue that this morpheme instantiates an *agentive v*/Voice, appearing with unergative, antipassive, and transitive (active) verbs in the language. This creates a paradigm of ways to create single argument structures in the language: for agent-argument intransitives, argumentless verbs combine with agentive *v*/Voice; for theme-argument intransitives, argumentless verbs either combine with directionals or one of three theme-introducing passive forms.

[16]This proposal entails abandoning a strict version Baker's (1985, 1988) Uniformity of Theta Assignment Hypothesis (UTAH), since themes are sometimes introduced in the specifier of *v*/Voice but sometimes introduced in the complement of Dir. While, as suggested by a reviewer, we could maintain UTAH by having Dir and *v*/Voice$_{\text{PASS}}$ instantiate the same head of *v*/VoiceP, such a syntax is difficult to motivate given word order facts in Mam. In particular, it would be difficult to explain why directionals do not occur in the same position as all other *v*/Voice morphemes. Note, however, that other relative versions of UTAH could be preserved; see e.g., Belletti and Rizzi 1988.

5.5 Summary

This section has provided an analysis of argument structure in Mam, with a focus on the introduction of internal arguments. In particular, we argued that in the active voice, directionals are necessary in transitive and unaccusative clauses because they are needed to introduce themes. In other words, internal arguments are severed from the argument structure of verbal roots, and their introduction is instead delegated to directionals. We then discussed voice alternations, concluding that both antipassives and (a subset of) passives circumvent the need for a directional. While the former fact was predicted by our theory (since antipassives demote internal arguments), the fact that directionals are not needed with three of four passive forms was unexpected. This led us to suggest that some flavors of passive *v*/Voice can also introduce themes.

In sum, we arrive at the following (tentative) inventory of argument-introducing functional heads in Mam:

(51) Theme-introducing
  a. **Directionals** (DIR)
  b. **Three flavors of passive *v*/Voice** (with existential closure of an agent/causer argument)

(52) Agent-introducing
  Agentive *v*/Voice

In the next and final section before the conclusion, we turn our attention to agentive *v*/Voice in (53). We argue that Mam makes its argument structure morphologically explicit: the morpheme *-n* appears if and only if an agent is present. We will thus propose that *-n* is the overt realization of an agentive *v*/Voice head, in parallel distribution with the passive *v*/Voice heads.

6 Putting it all together: Verbal functional structure in Mam

While the analysis in section 5 focused on the introduction of internal arguments, it was built on the assumption that Kratzer's (1996) severing of external arguments holds in Mam as well. In this section, we present evidence from the morphosyntax of Mam that this is in fact empirically motivated. In particular, section 6.1 argues that the morpheme *-n* is the overt manifestation of an agentive functional head (the counterpart of the passive heads discussed in the previous section). As previously mentioned, we adopt the analysis from Clemens and Coon (2018) and Coon (2019), which treats the relevant projection as a bundled *v*/Voice head (see also Harley 2017 for a similar approach).

Again in line with Clemens and Coon 2018 and Coon 2019, but also Elkins 2023 and Scott 2023, we will further propose in section 6.2 that an additional functional projection above *v*/Voice—realized as a "status suffix phrase" (SSP)—encodes the clause's valency. We will specifically argue that Mam can instantiate all functional morphemes distinctively in the morphology, sometimes at the same time. Finally, in section 6.3, we integrate the proposals from sections 5, 6.1, and 6.2 to present a comprehensive account of the extended verbal domain, capturing both morpheme distribution and surface word order.

The resulting theory ultimately paints a very neat picture of how verbal derivation is structured and realized in Mam, aligning with what can be described as a recent theoretical push towards the radical decomposition of argument structure (as in e.g., Castañeda 1967, Kratzer 1996, Borer 2005, Pylkkänen 2008, Ramchand 2008, Williams 2015, Levinson 2010, Elliott 2020, a.o.). The syntactic analysis likewise validates that our assumptions about external arguments are independently motivated in Mam.

6.1 The suffix *-n* as the head of an agentive *v*/VoiceP

In this section, we turn to the morpheme *-n* in Mam, arguing that it consistently instantiates the head of an "agentive" *v*/VoiceP. Specifically, it instantiates the agentive morpheme employed by Kratzer (1996) in (25):

(53) $[\![\text{-}n]\!] = \lambda x.\lambda e.\ \text{AGENT}(e) = x$

We argue that this head introduces agents regardless of transitivity, meaning that the subjects of unergatives, antipassives, and active transitives are all introduced by merging *-n* in *v*/VoiceP. Importantly, this reanalysis diverges from traditional descriptions of Mam, which have generally viewed the *-n* suffix in unergative and antipassive clauses as distinct from the *-'n* suffix found in transitive clauses (see e.g., England 1983b, 1988, 2017, Pérez and Jiménez 1997, Pérez Vail 2014, Elkins 2023, Scott 2023, and many more). Instead, we propose to decompose the suffix *-'n* into two segments: *-n* is the agentive suffix of (53) and is uniformly found in unergatives, antipassives and transitives, while *-'* is a transitive status suffix, only found in transitives (to be discussed in §6.2).[17]

Having put forward this proposal, we now show that it makes correct predictions. First, if *-n* has the semantics in (53), then it should only appear in clauses with agents, regardless of transitivity. This should encompass the following construction types: transitive (54a), antipassive (54b), and unergative (54c).[18] As shown in the examples below, this prediction is borne out. (Note that some of the examples in (54) are repeated from (43), but with "AGT" in the glosses reflecting our proposal in this section.)

(54) Agentive clauses with include *-n*

a. N=∅=o'k n-b'yo-'**-n**=i a=y ew.
IPFV=B2/3SG=DIR:in A1SG-hit-TV-AGT=DA DET=DA yesterday
'I was hitting you yesterday.' (transitive verb; SJA Mam; Scott 2023: 80)

b. N=chin b'yo-**n**=i t-i=y ew.
IPFV=B1SG hit-AGT=DA A2/3SG-RN:PAT=DA yesterday
'I was hitting you yesterday.' (antipassive verb; SJA Mam; Scott 2023: 80)

[17] More specifically, England (1983b) analyzes *-n* in intransitives as the antipassive suffix, and glosses it "AP" accordingly. For transitives, she observes that the morpheme *-n* is always accompanied by glottalization of the preceding vowel; she thus analyzes *-'n* as a single suffix that only appears on transitives. She also observes that transitives (nearly always) require directionals, and thus analyzes *-'n* as a "directional suffix" (DS). Most other work on Mam, including Scott 2023 and Elkins 2023, has adopted this analysis, a convention we have also assumed in glossed examples prior to this section. Under this kind of analysis, it is a coincidence that all agentive verbal constructions share *-n*. Here, we account for this shared property by analyzing it as the realization of agentive *v*, which we will gloss "AGT" in examples of this section.

[18] A notable exception to this pattern is the verb *b'et* 'walk'. Throughout the literature, this uneragtive intransitive appears without *-n*. However, in the author's field notes on SJA Mam, there are limited but existent examples of *b'et* occurring with *-n*.

c. Ma qo yol-**n**=i.
PROX B1PL speak-AGT=DA
'We talked.' (unergative verb; SJA Mam; Scott 2023: 266)

A second prediction concerns passives and unaccusatives, which are predicted not to merge an agentive *v*/Voice head. Again, this is borne out: as shown in (55), neither passives nor unaccusatives occur with *-n*:

(55) Passives and unaccusatives lack *-n*

a. Ma ∅ kub' qes-**j** n-q'ab'=i.
PROX B2/3SG DIR:down cut-PASS A2/3SG-hand=DA
'My hand was cut.' (passive; SJA Mam; Scott 2023: 83)

b. Ma tz'=e=tz **tz'aq** k'ab'il t-wi' mes.
PROX B2/3SG=DIR:out=DIR:come fall cup A2/3S-RN:top table
'The cup fell off the table.' (unaccusative; SJA Mam)

For passives (55a), we proposed in section 5.4.2 that passive *v*/Voice introduce the sole theme argument, and not an agent/causer (even though it existentially quantifies over an agent/causer). For underived unaccusatives (55b), we assume, following previous work (see e.g., Coon 2019, Oikonomou and Alexiadou 2022, Burukina and Polinsky 2025), that a separate unaccusative *v*/Voice head is merged; this head simply merges no DP in its specifier:

(56) $[\![\text{Ø}_{\text{UNAC}}]\!]$: $\lambda P_{\langle s,t\rangle}.\ P$

The unaccusative *v*/Voice head, as stated above, is a semantically vacuous identity function, which takes a property of events (the VP) and gives back the same property of events.

The crucial point, however, is that we can distinguish among two classes of verbs based on their ability to appear with *-n*. While transitive, unergative and antipassive constructions pattern together in taking *-n*, passives and unaccusative constructions pattern together in lacking *-n*. This is predicted if *-n* introduces agents, as proposed here. We therefore have an emerging typology of *v*/Voice heads in Mam, which vary in whether or not they introduce an argument, and if they do, what kind of argument they introduce (an agent or theme). The inventory of *v*/Voice heads in Mam is summarized in Table 7.

Table 7: Flavors of *v*/Voice in Mam

Head	Argument	Clause types	Morphology	Entry
AGT	agent	trans, unerg, antip	*-n*	(53)
PASS.1	theme	passive	*-j, -njtz, -Vt*	(49a-c)
PASS.2	—	passive	*-b'aj*	(49d)
UNAC	—	unaccusative	∅	(56)

Before moving to the next subsection, we discuss one cross-linguistic consequence of our approach regarding the syntactic status of antipassives. The current analysis of Mam "antipassives" does not formally distinguish them from unergatives: both simply combine with an agentive *v*/Voice head, instantiated by *-n*, and no theme is introduced. The same line of approach is recently proposed by Burukina and Polinsky (2025) for the Mayan language Kaqchikel concerning the cognate suffix *-Vn*. More specifically, Burukina and Polinsky show that there is only one way to derive an intransitive clause with one agent argument, encompassing both unergatives and antipassives. While this is not true for all Mayan languages (see Coon 2019 on Chuj), the shared syntax of unergatives and antipassives defended by Burukina and Polinsky extend to Mam. Thus, it is not the case that there is a special "antipassive head" that takes a transitive clause as its complement and manipulates a theme argument. Instead, antipassives arise from building an intransitive agentive clause with the suffix *-n*. Burukina and Polinsky go on to show a number of semantic diagnostics suggesting that the relevant Kaqchikel antipassives clearly do not involve an implicit theme, as expected if they are simply unergatives. While our preliminary investigation suggests that this is also the case in Mam, future work on Mam antipassives should verify that this is indeed the case. The current analysis predicts that it should be.

6.2 The status suffix phrase

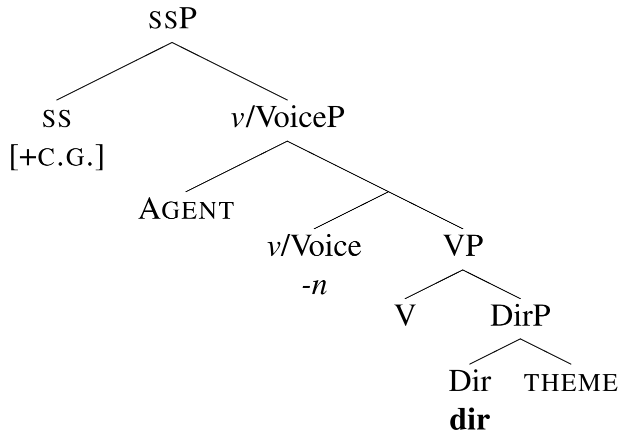
In Section 6.1, we proposed that agentive *v*/Voice maps neatly to Mam morphology via the suffix *-n*. Along with the passive morphology discussed in section 5.4.2, this now accounts for morphology within *v*/VoiceP, the domain of the extended VP which we proposed in section 5 is responsible for introducing arguments. We now expand our analysis, building on Clemens and Coon 2018 and Coon 2019, to include a "status suffix phrase" (SSP) projection in Mam. This projection will be useful for two reasons: (i) it will account for the systematic presence of a glottal stop with transitive verbs but not with intransitive verbs (unergatives and unaccusatives alike), and (ii) it will serve as the landing site for successive head movement necessary to derive Mam word order, which will be discussed in section 6.3.

Recall that Mam main verbs in *transitive* constructions always appear with glottalization on the final root vowel. That is, while they share with unaccusatives the requirement for directionals, and with unergatives the agentive suffix *-n*, transitive verbs are unique in systematically requiring glottalization. We thus propose that this glottal feature (formalized as a floating [+C(ONSTRICTED) G(LOTTIS)] feature) is the realization of the transitive "status suffix phrase (SSP)" head; we assume that intransitive SSP is not associated with any phonology (it is null).[19] We assume that status suffixes check the valency of the local verbal domain,[20] and that they occupy a position immediately above *v*/VoiceP:

[19]This is not the case in every Mayan language. In nearby Q'anjob'alan langauges, for instance, both transitive and intransitive status suffixes are overtly realized (see e.g., Mateo Toledo 2017 on Q'anjob'al and Royer, Mateo Pedro, Carolan, Coon, and Torres 2022 on Chuj). The [+CG] feature is moreover likely a cognate of the transitive status suffix in Q'anjob'alan languages. In Chuj, for instance, the transitive status suffix also involves glottalization (*-V'*), not found with the intransitive status suffix (*-i*).

[20]Delving into the details of how this is done is beyond the scope of this paper. One option, however, would be for SS to enter into Agree with the DPs it c-commands. Agreement with only one DP could yield -∅ (IV) while Agreement with more than one DP could yield [+C.G.] (TV). See e.g., Clem and Deal 2024 and Poole 2024 on how such a proposal, which relies on distinguishing one verus multiple steps of Agree, may be implemented.

(57) Functional structure for a transitive verb in Mam



As we will show in section 6.3, the linearization of this glottal morpheme is phonologically conditioned, necessarily docking on the final root vowel. For now, however, we motivate our proposal by showing that glottalization is indeed unique to transitives, occurring whenever there are two DP arguments in the extended verbal domain.

The glottal feature on the final root vowel is most clearly observed in comparing transitive verbal constructions with their antipassive counterparts, since both constructions have agents and thus -*n*. While the main differences in verbal morphology are that antipassives lack directionals and take Set B instead of Set A subject marking, we can also see that antipassives systematically lack final vowel glottalization. This was already illustrated in the examples above in (54a) and (54b) with the verb *b'yool* 'hit'. The two pairs of sentences below also illustrate the pattern once again. In the transitive (a) examples below, the glottal feature, which we gloss as "TV" for "transitive", is present. In the antipassive versions in (b), the glottal is absent.

(58) a. N=∅=o'k q-b'i**-'-n** ...
IPFV=B2/3SG=DIR:in A1PL-hear-TV-AGT ...
'We have heard that ....' (transitive; SJA Mam; Scott 2023: 289)

b. N=chin b'i**-n**=i.
IPFV=B1SG hear-AGT=DA
'I am listening.' (antipassive; SJA Mam)

(59) a. Junt, n=∅=xi t-chmo**-'-n**=i t-i'j.
next IPFV=B2/3SG=DIR:go A2/3SG-weave-TV-AGT=DA A2/3SG-RN:PAT
'Next, you start to weave it.' (transitive; SJA Mam; Scott 2023: 273)

b. N=qo chmo**-n**=i.
IPFV=B1SG weave-AGT=DA
'We are doing some weaving.' (antipassive; SJA Mam)

If [+C.G.] is the overt realization of a transitive status suffix, as proposed here, then we also expect it to be absent from passive sentences. This is indeed true in Mam: final root vowel glottalization is absent from passives. Compare the transitive imperative of *q'ama* 'say' in (60a) with the passive *q'ama-t* 'is told' in (60b).

(60) Passives lack [+C.G.] Voice (Scott 2023: 292,290)

a. Me'n t-xi t-q'ama**-'-n**=i.
NEG.V A2/3SG-DIR:go A2/3SG-say-TV-AGT=DA
"Don't say these words!" (transitive; SJA Mam)

b. ... n=∅=xi q'ama-t
... IPFV=B2/3SG=DIR:go say-PASS
'... she is told.' (antipassive; SJA Mam)

Similarly, we take the SSP projection to only be present in finite clauses; thus, we expect it to be absent from fully non-finite clauses in Mam. England (2013, 2017) describes a scale of finiteness in Mam and Scott (2023) proposes the syntax for four types of semi-/non-finite clauses. One type in particular is predicted to lack *v*/Voice (and thus also SSP): "fully non-finite" clauses, which lack aspect, directionals, and agreement, and appear with a non-finite suffix *-l* instead of the agentive *v*/Voice suffix *-n*. Crucially, these clauses also lack final vowel glottalization. These fully non-finite clauses appear as complements to verbs like *xi'* 'go' and *a'* 'start'.[21]

(61) Nonfinite clauses lack [+C.G.] (Scott 2023: 144)

a. Ma chj=e'x xjal [laq'o-l (t-e)].
PROX B2/3PL=go person [buy-NF A2/3SG-RN:PAT]
'The people went to buy (it).' (nonfinite; SJA Mam)

b. ... n=qw=a'=i [q'olb'e-l q=ib'=i].
... IPFV=B1PL=start=DISAGR [greet-NF A1PL-RN:RR=DA]
'... we started to greet each other.' (nonfinite; SJA Mam)

c. Ma tz'=ex xjaal [tx'ema-l si].
PROX B2/3SG=go person [cut-NF firewood
'The person went to cut wood.' (nonfinite; SJA Mam)

In sum, we have added an additional layer of functional structure to the extended verbal domain in Mam, namely the "status suffix phrase" (Clemens and Coon 2018). Unlike previous analyses, our approach captures the fact that the nasal suffix *-n* (AGT) is present in unergatives, antipassives, and transitives. It also explains why only transitive clauses have a glottal feature (SS.TV). Having made these proposals, the next and final subsection will present a comprehensive account of the elements we proposed for the extended verbal domain, which we show can capture both morpheme distribution and surface word order.

6.3 Functional structure, head movement, and morpheme order

So far, this paper has laid out an analysis of the syntax and semantics of the verbal functional elements responsible for argument structure and valency in Mam. First, we proposed that directionals are required for transitives and unaccusatives because they introduce theme arguments, which main verbs do not do. Second, we proposed that three of the four *v*/Voice passive morphemes offer an

[21]Nominals in fully nonfinite clauses can appear in one of two ways: (i) introduced by a relational noun, indicating oblique status, or (ii) as a pseudo-incorporated object of the verb, in which case we assume they are verbal modifiers, not verbal arguments (see Burukina and Polinsky 2025, Section 7 for discussion; they also cite Levin 2015).

alternative way to introduce themes, circumventing the need for a directional. Then, we proposed a morphological analysis of both agentive *v*/Voice and transitive SSP in the language. The relevant elements of the extended verbal projection are summarized in Table 8.

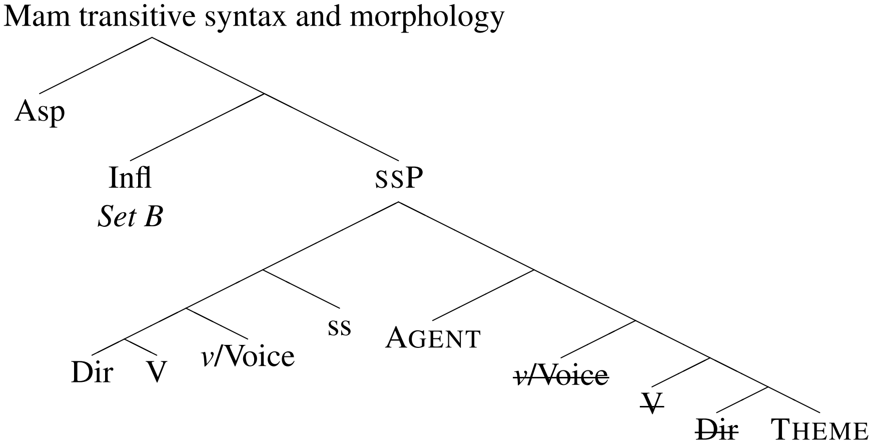
Table 8: Verbal functional sturcture in Mam *v*/Voice in Mam

Head	Argument	Clause types	Morphology	Entry
DIR	theme	trans, unac.	12 directionals	(28)
v/Voice.AGT	agent	trans, unerg, antip	*-n*	(53)
v/Voice.PASS.1	theme	passive	*-j, -njtz, -Vt*	(49a-c)
v/Voice.PASS.2	—	passive	*-b'aj*	(49d)
v/Voice.UNAC	—	unaccusative	∅	(56)
SS.TV	—	transitive	[+C.G.]	—
SS.IV	—	intransitive	∅	—

In this section, we now bring together these proposals by sketching the complete morphosyntax of the Mam extended verbal projection and argument domain. As already discussed in section 5, we adopt the head movement analysis of verbs in Mayan languages (Clemens and Coon 2018, Elkins 2023). Specifically, we adopt Clemens and Coon's (2018) analysis that the verb head moves to a landing site above the agent, to a head "at the edge of the extended verbal projection that sits above the subject". Like Clemens and Coon, we take this projection to be SSP.

For Mam, recall the clausal spine does not begin with the VP; verbs which need internal arguments select a DirectionalP (DirP). We assume the Dir head participates in head movement through V and *v*/Voice, landing at SS. Above SSP sit Infl and Asp, heads which Scott (2023) proposes are separate. Infl is responsible for object agreement (or default agreement in SJA Mam; see Scott (2023), chapter 3), while Asp provides aspectual information and morphology. The head movement analysis is illustrated below:

(62) Mam transitive syntax and morphology



An advantage of the analysis sketched in (62) is that the morpheme order that it derives is indeed the one observed for Mam, given in (63). Recall that Mam is VSO; aspect (and negation) delineate the left edge of the verbal complex, while status suffixes and *v*/Voice morphemes mark the right edge.[22] The subject and object follow the verb in that order.[23]

(63) [Asp - Set B - Dir - Set A - V - *v*/Voice - SS]$_{V}$ [AGENT]$_{S}$ [THEME]$_{O}$

While SS is linearly the final head in the verbal complex, which seems right given the position of status suffixes in other Mayan languages (see e.g., Clemens and Coon 2018), we argue that its realization is a floating [+C.G.] feature, and thus we do not expect it to appear categorically after the other verbal morphemes. Evidence that SS is the [+C.G.] feature and not a segmental /ʔ/ is that it is realized on the final root vowel, which may or may not be the final vowel of the verb. As shown in the examples in (64), in vowel-final verb roots, the final root vowel is glottalized.[24] This is true regardless of the shape/syllabic content of the root, as well as the presence and location of other glottalized consonants:

(64) [+C.G.] status suffix on V-final roots (Scott 2023: 273)

a. Tnel, n=∅=tzaj q-laq'o**-'**-n lan.
first, IPFV=B2/3SG=DIR:come A1PL-buy-TV-AGT wool.thread
'First, we buy wool thread.' (SJA Mam)

b. Junt, n=∅=jaw q-q'ano**-'**-n.
next IPFV=B2/3SG=DIR:up A1PL-organize.thread-TV-AGT
'Next, we organize the thread.' (SJA Mam)

c. ... n=∅=xi t-kyji**-'**-n=i t-txa'n.
... IPFV=B2/3SG=DIR:go A2/3SG-twist-TV-AGT=DA A2/3SG-nose
'... you twist the ends.' (SJA Mam)

With consonant-final verb roots, the agentive *v*/Voice suffix has two allomorphs. Before the disagreement enclitic *=i*, it appears as *-n*, shown in (65a). When *-n* is word-final following a root consonant, a schwa-like vowel is epenthesized, represented orthographically as <a>. This epenthetic vowel does not attract glottalization; instead, the final vowel of the root is glottalized, shown in (65b) and (65c).

(65) [+c.g.] Voice on on C-final roots

a. Ma tz'=ok q-ke'y-**n**=i a=y.
PROX B2/3SG=DIR:in A1PL-see-AGT=DA DET=DA
'We saw you.' (SJA Mam)

[22] Coon (2017, 2019) argues, for the Mayan languages Ch'ol and Chuj respectively, that ergative "Set A" morphemes are agreement prefixed to the verb root and originate from *v*/Voice. Different analyses of Set A assignment would be compatible with the structure we propose (see e.g., Deal and Royer 2025).

[23] Royer (2022, 2025) shows direct evidence that objects raise above subject in Mam. To derive VSO word order with high objects, we follow Scott (2023), who builds on a body of literature on VSO in Mayan in assuming rightward specifiers for the subject and the object's landing site in Mam (see also Little 2020). This is not indicated in the tree for brevity. Additionally, we adopt Scott's analysis of the disagreement enclitic *=i* as the *in-situ* partial spell out of certain pronominal arguments. When the disagreement enclitic references the subject, it appears immediately to the right of the verbal complex, correctly predicted by the present analysis.

[24] /ʔ/ does not appear to have phonemic status in Mam: it is often epenthesized to fill out a minimal CVC root requirement. The realization of glottal epenthesis differs between dialects. In Todos Santos Mam, for example, Elkins and Kuo (2022) demonstrate that in word-final position, a consonantal /ʔ/ is pronounced, and additionally undergoes a process of aspiration found on all word-final voiceless stops. After a vowel but before any additional coda consonants, /ʔ/ is realized as creaky phonation, increased phonetic duration, and a falling tonal contour on the vowel.

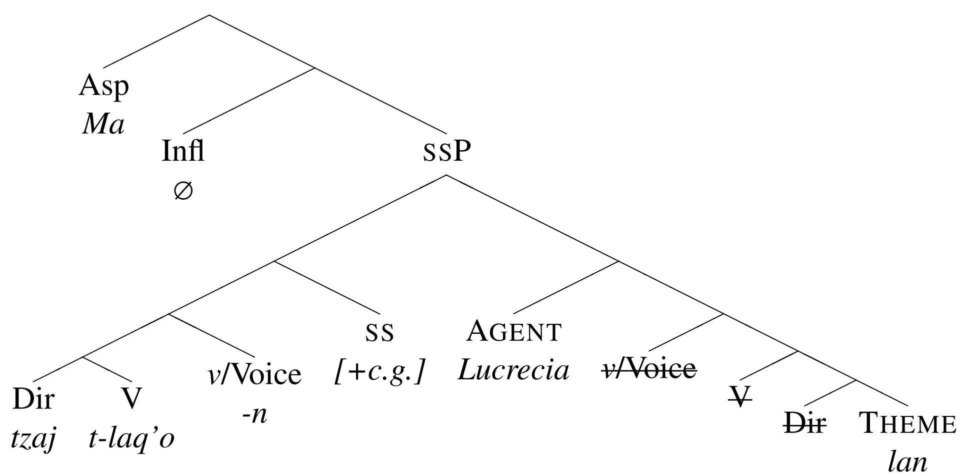
b. Ma tz'=ok ky-ke'y-**an** qa qin=i.
PROX B2/3SG=DIR:in A2/3PL-see-AGT PL 1SG=DA
'They saw me.' (SJA Mam)

c. Ma ∅ jaw q-xk'lo'x-**an**.
PROX B2/3SG DIR:up A1PL-wrap-AGT
'We wrapped it.' (SJA Mam)

With all pieces of the analysis in place, we can now understand (i) how each argument in the Mam verbal clause is introduced, (ii) how the functional heads map to morphology, and (iii) how morpheme order is derived via head movement. A transitive clause is given in (66), and a tree illustrating our final proposal for the structure of the extended verbal domain is given in (67).

(66) Ma ∅ tzaj t-laq'o-'-n Lucrecia lan.
PROX B2/3SG DIR:come A2/3SG-buy-TV-*v*.AGT Lucrecia wool.thread
'Lucrecia bought wool thread.' (SJA Mam)

(67) Mam transitive clause (tree for 66)



7 Conclusion

In this paper, we proposed that directionals in Mam have a dedicated semantic function that goes beyond that of spatio-temporal deixis: they play a fundamental role for argument structure. We showed that there is reason to believe that the denotation of Mam verbs ought to be severed not just from their external arguments (à la Schein 1993 and Kratzer 1996), but also from their *internal* arguments; with verbs constituting properties of events, directionals are required to introduce themes. This proposal makes it possible to account for the observation that transitives and unaccusatives require directionals, whereas unergatives do not.

We then showed how this analysis can be brought to bear on voice constructions such as the passive and antipassive, in which directionals are not required. Whereas the behavior of antipassives (in which internal arguments are absent or demoted to oblique status) follows straightforwardly from our approach, the behavior of three of the four passive morphemes in Mam required some care. We adopted a novel theory of these passives for Mam in which the subject is not the internal argument, but is instead a thematic subject introduced by a passive *v*/Voice morpheme. Given Mam's rich inventory of voice morphology, the analysis of passives in this paper (and of grammatical voice in Mam in general) is only preliminary.

Turning from the implications of directionals as introducers of themes, we presented evidence that the introduction of agents may also be discovered from the verbal morphology of Mam. In particular, following Clemens and Coon (2018), we proposed to split the thematic domain into *v*/VoiceP and SSP, the latter of which is only overtly instantiated in transitives. An agentive *v*/VoiceP is realized by the suffix *-n*, and is therefore also present on unergative/antipassive verbs, which are not formally differentiated in the system we proposed (as is the case in other work on Mayan languages; Burukina and Polinsky 2025). Transitives are then differentiated from intransitives insofar as they bear a transitive status suffix, realized by a floating [+C.G.] feature, which docks on the final root vowel.

Importantly, the argument for the severance of arguments in Mam in this paper was based solely on the observed mapping between morphosyntax and argument structure: since a directional arises whenever a theme is present, we inferred that the directional is required to introduce the theme relation. While this constitutes one argument in favour of severing thematic relations from the verb's meaning, it only constitutes one type of evidence for our approach. As pointed out by an anyonymous reviewer, other types of empirical evidence—perhaps even stronger than the one introduced in this work—should be undertaken in future work. For instance, severing approaches to argument structure predict empirical effects with regards to an arguments' semantic autonomy in relation to the verb's meaning. One way to test for this is with intervening operators, such as modals. As discussed in Williams 2015, based on a previous version of a now published manuscript by Schein (2017), the distribution of adnominal modals can serve as another argument for separating thematic relations from verbs. Indeed, a sentence like (68a) expresses the claim in (68b).[25]

(68) a. The cops gathered the students and possibly the professors. (Williams 2015: 193)
b. There are two possibilities: either (i) the cops gathered the students or (ii) the cops gathered the students and the professors.

What is the scope of the modal in the English sentence in (68a)? It is neither (69a) nor (69b). Instead, Williams proposes that the adnominal modal must only operate on the theme predicate, as illustrated in (69c).

(69) a. (68b) ≠ there is an event *e*; $\Diamond$[*e* is a gathering of the students and the professor]; the agents are the cops in *e*.
b. (68b) ≠ there is an event *e*; *e* is gathering of the students and $\Diamond$[*e* is a gathering of the professors]; the cops are agents in *e*.
c. (69c) = there is an event *e*; *e* is a gathering; the cops are agents in *e*; the students are themes in *e*; $\Diamond$[the professors are themes in *e*]

[25] This diagnostic nevertheless presents a potential confound, insofar as the sentence "*The cops gathered possibly the professors" is ill-formed by itself. Thus, it could be that the test sentence involves some form of ellipsis: "The cops gathered the students and possibly <gathered> the professors". If an elided verb is necessarily syntactically present and interpreted, a reading like the one in (69c) is actually expected, as two separate events are introduced.

Crucially, a meaning like (69c) is only possible if the theme is introduced independently. Providing careful semantic evidence for the severing of thematic relations in Mam, as done for English here, would provide additional support for our approach. Recent work on semantic evidence for theme relation separation from another underdocumented language, Kanien'kéha (Myers 2025b), shows that this strand of research has significant potential. For an overview of potential empirical evidence in favour of severing arguments, see Williams 2015, chapter 9.

In conclusion, we find that identifying a dedicated function for directionals for argument structure in Mam opens up a number of avenues for investigating its thematic domain more generally. Seeing as directionals in other Mayan languages have also recently been implicated in the derivation of certain verbal constructions (e.g., indirect arguments (Mateo Toledo 2004, 2008, 2023) and stative predication (Henderson et al. 2018, Elias 2019) in Q'anjob'alan languages), we hope that this work ushers in further fruitful research on directionals across the Mayan language family.

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